



Clay and organic matter interaction in the biogeochemical genesis of alkaline soda-lake water

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Abstract

This controlled experimental study examines bentonite, compost, and their combination (bentonite+compost) as biogeochemical drivers in the genesis of alkaline soda-lake water. Treatments affected mineral–water reactions (including hydrolysis), ion partitioning, and nutrient cycling processes, controlling brine evolution under non-flow-through, aerated bottle conditions in the laboratory. pH ranged from 8.18 to 10.35, highest in bentonite-only treatments, while electrical conductivity peaked in bentonite+compost at 995 $\mu\text{S cm}^{-1}$. At the final sampling point (day 30), the bentonite+compost treatment showed the strongest Na enrichment and carbonate-system response, with Na^+ reaching 61.64 mg L^{-1} and the carbonate-species signal 12.83 mg L^{-1} ($\text{HCO}_3^- + \text{CO}_3^{2-}$), compared with 52.53 mg L^{-1} Na^+ and 7.12 mg L^{-1} carbonate-species signal in bentonite-only and 12.77 mg L^{-1} Na^+ and 4.93 mg L^{-1} carbonate-species signal in compost-only. Compost-only produced the highest K^+ (162.11 mg L^{-1}), Ca^{2+} (46.80 mg L^{-1}), and Mg^{2+} (20.39 mg L^{-1}), whereas bentonite-only showed the lowest dissolved Ca^{2+} (3.30 mg L^{-1}) and Mg^{2+} (2.26 mg L^{-1}); bentonite+compost was intermediate for divalent cations (Ca^{2+} 14.36 mg L^{-1} , Mg^{2+} 5.54 mg L^{-1}) and K^+ (67.38 mg L^{-1}). For conservative anions, Cl^- increased across treatments from bentonite-only (241.02 mg L^{-1}) to compost-only (329.20 mg L^{-1}) to bentonite+compost (446.78 mg L^{-1}), while SO_4^{2-} was lowest in bentonite+compost (5.87 mg L^{-1}) relative to bentonite-only (23.71 mg L^{-1}) and compost-only (29.30 mg L^{-1}). Compost-treated samples showed nutrient enrichment, with TN increasing from 13.54 to 28.47 mg L^{-1} , TP from 7334.33 to 13475.5 mg L^{-1} , and TOC from 67.01 to 174.93 mg L^{-1} , whereas bentonite contributed minimal nutrients but maintained higher and less variable pH. Evaporation rates differed among treatments (20.77, 18.39, and 17.95 mL day^{-1} for compost-only, bentonite-only, and bentonite+compost, respectively). The results are interpreted primarily from the solute chemistry dataset, with solid-phase and microbial effects inferred rather than directly measured. This deeper understanding of the biogeochemical genesis of alkaline soda waters provides crucial practical insights for the effective management and ecological restoration of these unique ecosystems.

Keywords Bentonite · Compost · Alkaline soda-lake water · Hydrolysis · Ion partitioning · Brine evolution · Bentonite+compost

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Introduction

Soda lakes (also termed alkaline lakes) are unique aquatic ecosystems characterized by permanently high pH (typically > 9) and ionic dominance (on an equivalents basis) by Na^+ together with carbonate-system species ($\text{HCO}_3^- + \text{CO}_3^{2-}$) (Boros and Kolpakova 2018). These lakes occur predominantly in arid and semi-arid regions worldwide, including the East African Rift Valley, Eurasia, and parts of North and South America (Kempe and Kazmierczak 2011; Boros and Kolpakova 2018). Their formation reflects coupled geological and climatic controls in endorheic basins where evaporation exceeds outflow and sodium enrichment

proceeds relative to Ca^{2+} and Mg^{2+} (Ross et al. 2014; Pecoraino et al. 2015).

These systems are characterized by high salinity and alkalinity, with Na^+ and carbonate-system species dominating the major-ion composition. Despite these extreme conditions, adapted microbial communities occur and microbial metabolisms can generate or consume alkalinity in lake water and sediment pore water, influencing carbonate-system chemistry and nutrient cycling (Grant 2004; Boros and Kolpakova 2018; Haines et al. 2023).

Understanding the chemical dynamics and ecological roles of soda-lake waters requires considering solid–water interactions at the sediment–water interface and on suspended particles, where surface reactions can modify dissolved-ion composition. Here, “distribution” refers to partitioning of ions between dissolved water and solid-associated pools (sediments/particles), affecting nutrient and trace-element availability and water chemistry (Tóth 1999).

Conditions required for natural soda-lake development include an underfilled tectonic basin, where subsidence creates accommodation faster than sediment infill, and closed hydrography in which evaporation exceeds outflow. Under these conditions, progressive chemical evolution favors high Na^+ relative to Ca^{2+} and Mg^{2+} , and carbonate-system species can become dominant over sulfate and chloride in alkaline brines (Tosca and Tutolo 2023).

Bentonite is a clay material primarily rich in montmorillonite. Montmorillonite-rich (smectite-rich) clay minerals commonly occur in the sediments and catchments of many alkaline soda lakes and closed-basin environments, including the soda pans of the Carpathian Basin, the East African Rift Valley, and other volcanic or weathered sedimentary settings, where they play an important role in controlling ion exchange and water chemistry (Deocampo and Jones 2013). Its high cation exchange capacity (CEC) and swelling characteristics allow bentonite to influence ion retention and solution chemistry through solid–water interactions, including surface adsorption and exchange-related processes, in aqueous systems. In this study, bentonite is used as a model mineral phase in a bottle-water system. The composition and treatment of bentonite affect its exchange behavior; for example, sodium activation can increase CEC relative to raw bentonite (Bahranowski et al. 2021).

Compost can substantially alter the chemistry of saline–alkaline aquatic systems by releasing dissolved organic matter and nutrients (particularly nitrogen- and phosphorus-containing compounds) during decomposition (Bernal et al. 2009). In natural soda-lake and saline-wetland settings, comparable decomposable organic-matter inputs may occur through livestock droppings in grazed wetlands, decomposition of aquatic and shoreline vegetation, and runoff from surrounding agricultural areas where manure- or

compost-based fertilizers are applied. Compost can also provide a substrate and inoculum for microbial activity (Haines et al. 2023). In a non-flow-through aerated bottle system such as the present experiment, these processes are expected to increase dissolved nutrient and organic-carbon concentrations and to modify carbonate-system behavior, pH, and ion partitioning. Accordingly, in this study compost is treated as a model organic-matter input that drives chemical and microbially influenced changes in the solute phase (Bernal et al. 2009).

This experiment is an aqueous bottle model (water + solids with continuous aeration) rather than a soil study. Accordingly, interpretations are limited to water-column solute chemistry and solid–water interactions within the bottle system.

The genesis of alkaline brines in soda lakes is controlled by interacting geological, hydrological, climatic, physico-chemical, and biological factors, including basin setting, evaporation–inflow balance, temperature, atmospheric CO_2 exchange, and water–sediment/mineral interactions (Tosca and Tutolo 2023). In this context, organic matter and reactive clay minerals may influence carbonate-system chemistry, mineral precipitation/dissolution, and ion partitioning between dissolved and solid-associated phases, thereby contributing to the chemical evolution of lake water toward alkaline brine conditions (Kleber et al. 2021).

Alkaline soda lakes are globally unique and threatened ecosystems with high conservation priority, including within the Natura 2000 network of the European Union (Boros and Vörös 2024). In protected soda-lake wetlands, controlled grazing by domestic livestock is used to maintain open wetland surfaces (Boros et al. 2013); however, the biogeochemical effect of livestock droppings on soda-lake water chemistry remains insufficiently quantified (Boda et al. 2021).

Therefore, our goal was to compare how bentonite and compost affect solute chemistry and alkaline soda-lake water related brine evolution under controlled laboratory conditions using three treatments: bentonite-only (model mineral/clay phase), compost-only (model organic-matter input), and bentonite+compost (combined mineral–organic treatment). This comparison is important because it separates mineral-driven, organic-matter-driven, and combined mineral–organic effects on soda-lake-relevant water chemistry, providing mechanistic insight into processes that are difficult to isolate directly in natural wetlands.

Working hypotheses were that (H1) bentonite increases and stabilizes alkaline conditions relative to compost-only and the tap-water baseline, (H2) bentonite+compost produces stronger Na^+ and carbonate-species dominance ($\text{HCO}_3^- + \text{CO}_3^{2-}$) than either component alone, and (H3) compost increases dissolved divalent cations while

bentonite presence reduces their dissolved fractions. We expected bentonite to yield higher and less variable pH, bentonite+compost to show the strongest alkaline soda-lake water like ionic response, and compost-only to produce the highest nutrient and TOC release. Based on field observations from grazed wetlands, we further hypothesized that the combined bentonite+compost treatment would produce stronger alkaline soda-lake water related chemical responses than either single-component treatment.

Materials and methods

Experimental design and laboratory analysis

The experiment was conducted from 11 November 2024 to 10 December 2024 (four-week incubation) at the Hungarian University of Agriculture and Life Sciences, Gödöllő, Hungary. This duration was selected as a short-term incubation window to capture early treatment-driven changes in solute chemistry relevant to brine evolution under controlled laboratory conditions. Sodium bentonite powder (vendor: Unikén Kft., Kecel, Hungary) and a commercial, vendor-certified compost product (manufacturer/supplier: Profikomp Zrt., Gödöllő, Hungary) were commercially sourced. Bentonite was oven-dried at 200 °C for 3 h as a laboratory pre-treatment to reduce variability in initial water content and standardize the starting material across replicates. It was applied as a fine powder fraction (20–80 µm) to provide reactive clay surfaces and reproducible mineral–water contact in the bottle system; the supplier label specifies an active SiO₂ content of ≥ 65–70%. Compost was homogenized with a mortar and used at a coarser grain-size fraction (~0.5–3 mm) to preserve representative particulate organic matter while reducing sample heterogeneity. The different size fractions reflect the different physical nature and intended function of the two amendments (reactive mineral phase vs. particulate organic-matter input). According to the accredited analytical certificate (BÁLINT Analitika Kft., Budapest; laboratory report for compost samples, client: Profikomp Zrt.), the compost (dry-weight basis) had pH(H₂O) 7.29–7.45, organic matter 48.8–51.4%, total N 1.34–1.42%, P₂O₅ 1.24–1.60%, and K₂O 2.07–2.25% (with Ca 5.30–5.56% and Mg 0.71–0.82%). The certificate also reported particle-size compliance (<20 mm: 100%) for the submitted compost samples.

Bentonite and compost comprise multiple material fractions (solid matrix, exchangeable ions, and water-extractable components). Accordingly, this study reports solute-phase chemistry in the 0.45 µm-filtered bottle water and starting material specifications are documented using supplier labels and the compost certificate. Exchange-phase composition

was not directly measured; therefore, exchange-related interpretations are presented as inferences from solute patterns and treated as mechanistic hypotheses rather than direct measurements.

A completely randomized design was employed with three treatments: (1) bentonite-only (65 g), (2) compost-only (15 g), and (3) bentonite+compost (65 g bentonite + 15 g compost). The bentonite and compost amounts were selected to generate measurable treatment-driven changes in solute chemistry within the four-week bottle experiment while preserving their different experimental functions (bentonite as the reactive mineral phase, and compost as the organic-matter input and microbial substrate). Each treatment was replicated three times (nine experimental units total). Bentonite was used as a representative clay-rich mineral substrate relevant to clay-bearing, low-permeability layers in wetland settings, while compost was used as a representative organic-matter input. Tap water was selected as a reproducible electrolyte matrix with Ca–Mg–HCO₃-type background chemistry common in many non-saline freshwaters.

All treatments were established in 5-L glass bottles filled with tap water and maintained in a dark environment at ambient laboratory temperature (~25 ± 2 °C). The same bottle type and filling volume were used for all replicates; therefore, exposed water surface and headspace conditions were comparable among treatments. Each bottle was continuously aerated at a low rate to maintain oxic conditions and to provide bubble-induced gentle mixing (weak circulation without vigorous turbulence or mechanical stirring), sufficient to limit stratification while allowing settling of the heavier solid fraction. Aeration was supplied using an aquarium air pump via open-ended tubing, with one air line per bottle operated at a low bubbling rate. The bottles were open to gas exchange with laboratory air but non-flow-through (no drainage/outflow). Room humidity was not actively controlled; however, all bottles were incubated under the same laboratory conditions. Evaporation was quantified from daily water-level measurements, and evaporative losses were replenished with deionized water to maintain constant volume. Initial (0 h) samples were collected immediately after treatment setup (addition of bentonite and/or compost to water), followed by 0.45 µm filtration and analysis; the same analyses were repeated at the end of the experiment.

Municipal tap water was used to provide a reproducible background electrolyte matrix for the bottle experiment. A composite sample of the same batch was analyzed before treatments (25 °C ± 2 °C): pH = 7.5; EC = 565 µS cm⁻¹; Na⁺ = 13.5 mg L⁻¹; K⁺ = 1.48 mg L⁻¹; Ca²⁺ = 67.77 mg L⁻¹; Mg²⁺ = 25.9 mg L⁻¹; HCO₃⁻ = 2.70 mg L⁻¹; Cl⁻ = 28.2 mg L⁻¹; SO₄²⁻ = 20.2 mg L⁻¹.

Throughout the experimental period, pH and electrical conductivity (EC) were measured daily using a pH/EC meter (HANNA HI-9821). Major cations were analyzed using inductively coupled plasma mass spectrometry (ICP-MS; Shimadzu Corporation, Kyoto, Japan). Anion concentrations were determined according to the standard procedures outlined by the American Public Health Association (APHA, 2005). Measurements were recorded on 24 sampling days across the four-week incubation, with no measurements taken on weekends.

Carbonate-system species (HCO_3^- and CO_3^{2-}) were determined by acid–base titration (alkalinity titration) following APHA standard methods [15]. Briefly, filtered samples were titrated with standardized strong acid to the relevant alkalinity endpoints, and bicarbonate and carbonate contributions were quantified from alkalinity in combination with the measured pH. Species partitioning was calculated using standard carbonate alkalinity relationships at $\sim 25^\circ\text{C}$ ($\pm 2^\circ\text{C}$). Chloride (Cl^-) was determined by argentometric titration following APHA standard methods (APHA, 2005), using standardized AgNO_3 as titrant (with Cl^- quantified from the formation of AgCl at the endpoint). Sulfate (SO_4^{2-}) was measured by spectrophotometric determination following APHA standard methods (APHA (American Public Health Association) 2005), using a turbidity-based (BaSO_4) procedure and quantification against appropriate standards.

Total nitrogen (TN) and total phosphorus (TP) concentrations were measured colorimetrically using a NANO-COLOR spectrophotometer (Macherey–Nagel GmbH & Co. KG, Germany). Total organic carbon (TOC) in the water samples was measured using a TOC analyzer (Shimadzu TOC-L or equivalent).

For classification in this study (at $\sim 25 \pm 2^\circ\text{C}$), a sample was considered alkaline soda-lake water if it met all of the following criteria: (i) $\text{pH} \geq 9.0$; (ii) Na^+ dominance among cations on an equivalents basis ($\geq 50\%$ of total measured cations); and (iii) dominance of carbonate-system species ($\text{HCO}_3^- + \text{CO}_3^{2-}$) among anions on an equivalents basis ($\geq 50\%$ of total measured anions). Ion dominance was evaluated using equivalents-percent (e%), calculated after converting ion concentrations from mg L^{-1} to meq L^{-1} with ion-specific equivalent weights; thus, dominance was assessed on a charge-equivalent basis rather than mass

concentration, consistent with soda-lake biogeochemical classification practice (Boros and Kolpakova 2018).

Data analysis

The statistical analysis was carried out in OriginPro 2024 (OriginLab, Northampton, MA, <http://www.originlab.com>), and graphical data were visualized using GraphPad Prism version 10 (GraphPad Prism version 10.0.0 for Windows, GraphPad Software, Boston, Massachusetts, USA, www.graphpad.com). The normality of variables was assessed using the Kolmogorov–Smirnov test, and log transformation was applied where necessary. Treatment effects and temporal variations were evaluated using the measured water-chemistry dataset (daily pH and EC; major ions and nutrients at defined sampling points) and analyzed using a two-way multivariate analysis of variance (MANOVA), with treatment and experimental phase as fixed factors, including their interaction term. MANOVA was selected because multiple correlated response variables were evaluated simultaneously within this experimental design. To protect against inflating Type I error rates, significant multivariate effects were followed up with univariate two-way analyses of variance (ANOVAs) for each dependent variable. Pairwise comparisons were subsequently performed using Tukey's post-hoc tests where applicable. Statistical significance for all tests was established at $\alpha = 0.05$. Exchange-related interpretations are based on solute-phase patterns rather than direct measurements of the solid exchange phase.

Results

Daily measurements of the treatments

The multivariate analysis of variance (MANOVA) revealed highly significant main effects for both treatment and phase, as well as a significant treatment $\times$ phase interaction on the examined water quality parameters (Table 1).

Detailing the changes during the experiment, the Fig. 1 illustrates the temporal trends of pH and EC across three treatments over the experiment period. The pH levels ranged from 8.18 to 10.35, with the highest values observed in the treatment containing only bentonite (10.35 to 9.26).

Table 1 Results of the multivariate analysis of variance (MANOVA) testing the effects of treatment, phase, and their interaction on the examined water quality parameters

Source	Wilks' Lambda	F	df _{hyp}	df _{error}	p	
Treatment	0.002	$F_{(18;44)} = 60.12$	18	44	< 0.001	0.961
Phase	0.042	$F_{(9;22)} = 55.30$	9	22	< 0.001	0.958
Treatment $\times$ Phase	0.011	$F_{(18;44)} = 20.90$	18	44	< 0.001	0.895

(***) = $p < 0.001$; (**) = $p < 0.01$; (*) = $p < 0.05$; (ns) = $p \geq 0.05$

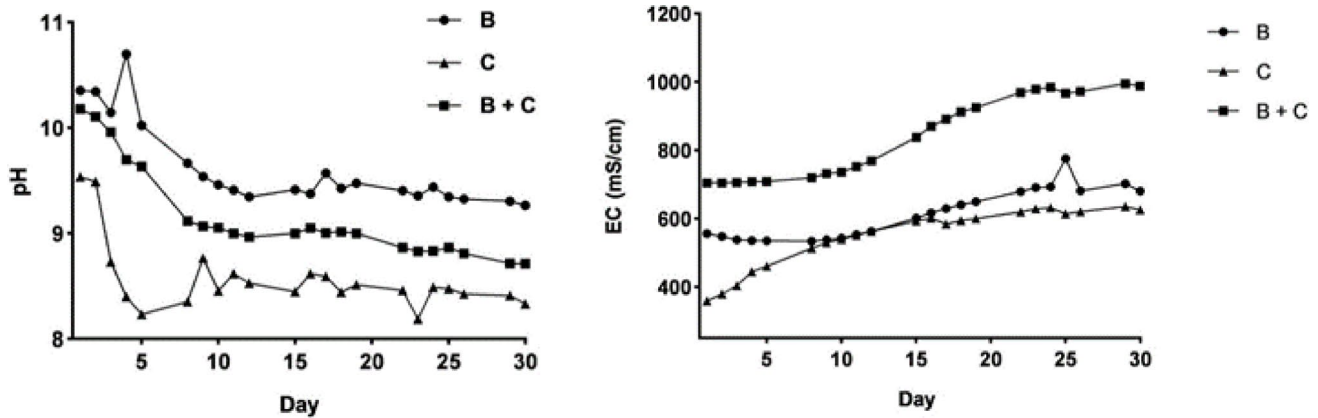


Fig. 1 Changes in pH and EC over the course of the experiment; statistic shown: mean $\pm$ SEM ($n = 3$ per treatment at each time point); matrix: bulk water; EC reported as $\mu\text{S cm}^{-1}$ ($25^\circ\text{C} \pm 2^\circ\text{C}$)

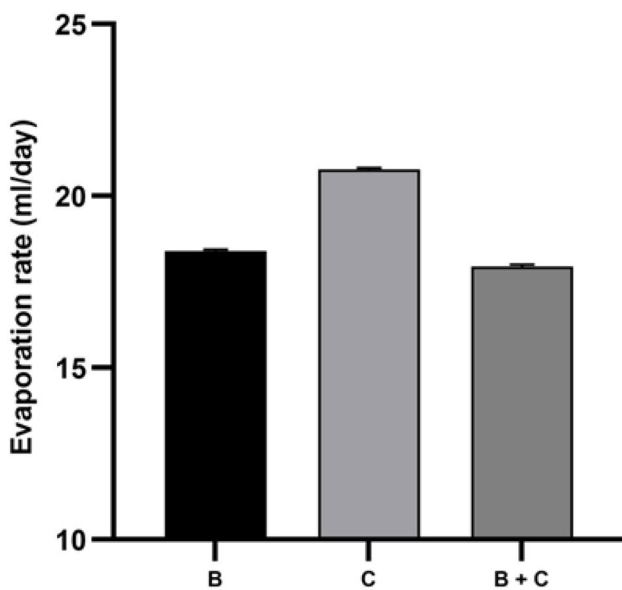


Fig. 2 Evaporation rate observed during the experiment; statistic shown: mean $\pm$ SEM ($n = 3$ per treatment); units: mL day^{-1} ; treatment means (mL day^{-1}): bentonite [18.39 ± 0.05], compost [20.77 ± 0.05], bentonite+compost [17.95 ± 0.05]

Conversely, the lowest pH readings (9.53 to 8.18) were recorded in the compost-only treatment. The EC value ranged from $359 \mu\text{S cm}^{-1}$ to $995 \mu\text{S cm}^{-1}$. The combined treatment (bentonite+compost) showed the highest EC values ($704\text{--}995 \mu\text{S cm}^{-1}$), while the compost-only treatment had the lowest ($359\text{--}635.66 \mu\text{S cm}^{-1}$).

Samples with compost alone had a higher evaporation rate ($20.77 \text{ mL day}^{-1}$) than those with bentonite ($18.39 \text{ mL day}^{-1}$) or bentonite+compost ($17.95 \text{ mL day}^{-1}$) (Fig. 2).

Composition changes in the treatments

Follow-up univariate ANOVAs showed that all examined water quality parameters were significantly affected by the treatment (Table 2).

Detailing the changes during the experiment, at 0 h, Na^+ concentrations were 16.46 mg L^{-1} in the compost-only treatment, 37.88 mg L^{-1} in the bentonite-only treatment, and 40.80 mg L^{-1} in the bentonite+compost treatment. Na^+ was higher in bentonite-only than compost-only ($p = 0.014$), and higher in bentonite+compost than compost-only ($p < 0.01$), whereas bentonite-only and bentonite+compost did not differ significantly (n.s.) (Fig. 3). By day 30, Na^+

Table 2 Summary of the follow-up univariate between-subjects effects (ANOVA) for each independent water quality variable

Variable	Treatment	Phase	Treatment $\times$ Phase
Na	$F_{(2,30)}=42.65$ (***)	$F_{(1,30)}=9.33$ (**)	$F_{(2,30)}=4.50$ (*)
Carbonates	$F_{(2,30)}=30.17$ (***)	$F_{(1,30)}=0.01$ (ns)	$F_{(2,30)}=0.92$ (ns)
TOC	$F_{(2,30)}=69.73$ (***)	$F_{(1,30)}=39.53$ (***)	$F_{(2,30)}=14.50$ (***)
TN	$F_{(2,30)}=116.09$ (***)	$F_{(1,30)}=33.68$ (***)	$F_{(2,30)}=22.22$ (***)
TP	$F_{(2,30)}=35.52$ (***)	$F_{(1,30)}=63.35$ (***)	$F_{(2,30)}=10.51$ (***)
K	$F_{(2,30)}=363.88$ (***)	$F_{(1,30)}=89.50$ (***)	$F_{(2,30)}=51.91$ (***)
Ca	$F_{(2,30)}=14.25$ (***)	$F_{(1,30)}=0.55$ (ns)	$F_{(2,30)}=0.42$ (ns)
Mg	$F_{(2,30)}=6.28$ (**)	$F_{(1,30)}=0.07$ (ns)	$F_{(2,30)}=0.21$ (ns)
Cl	$F_{(2,30)}=10.70$ (***)	$F_{(1,30)}=10.92$ (**)	$F_{(2,30)}=1.85$ (ns)

(***) = $p < 0.001$; (**) = $p < 0.01$; (*) = $p < 0.05$; (ns) = $p \geq 0.05$

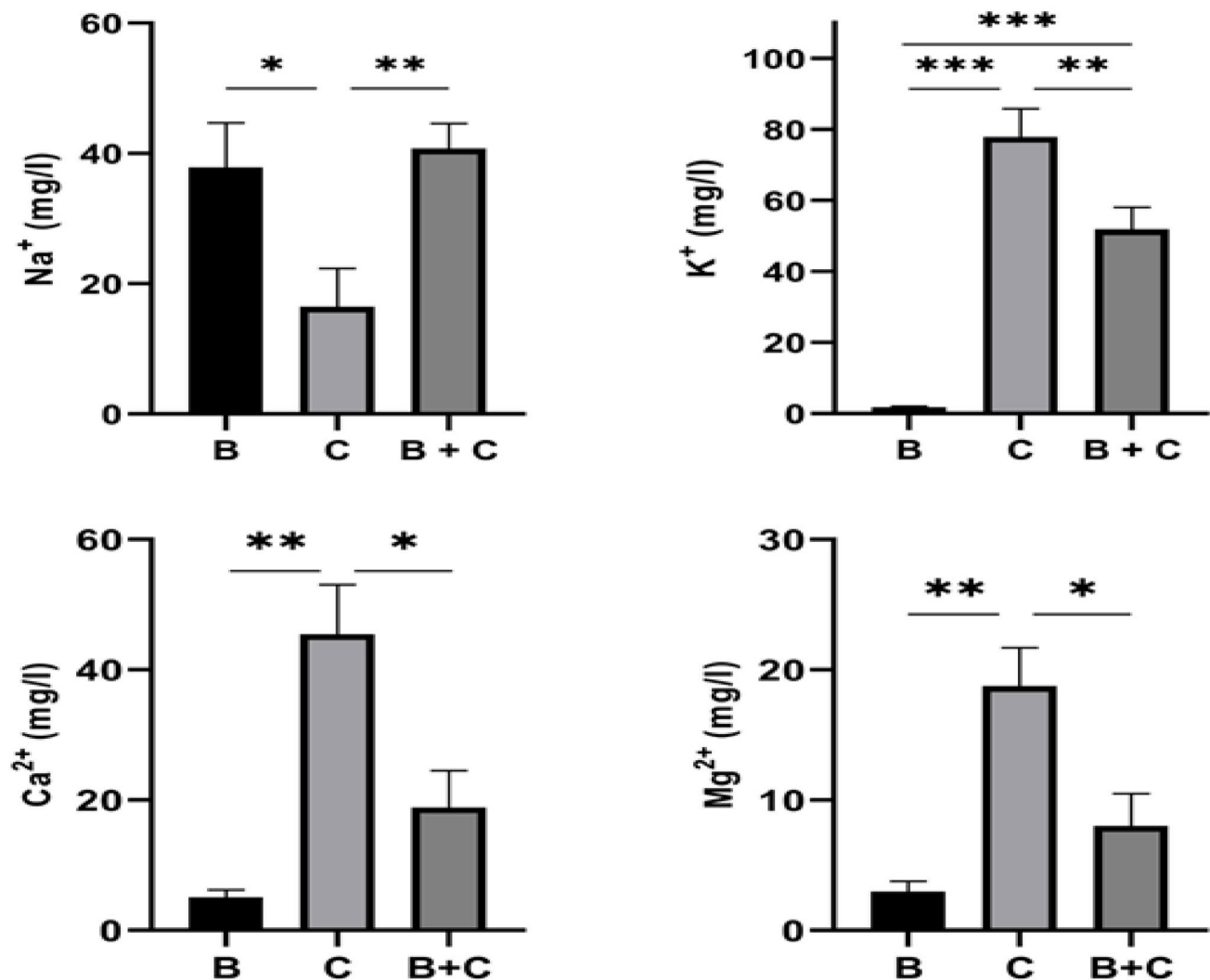


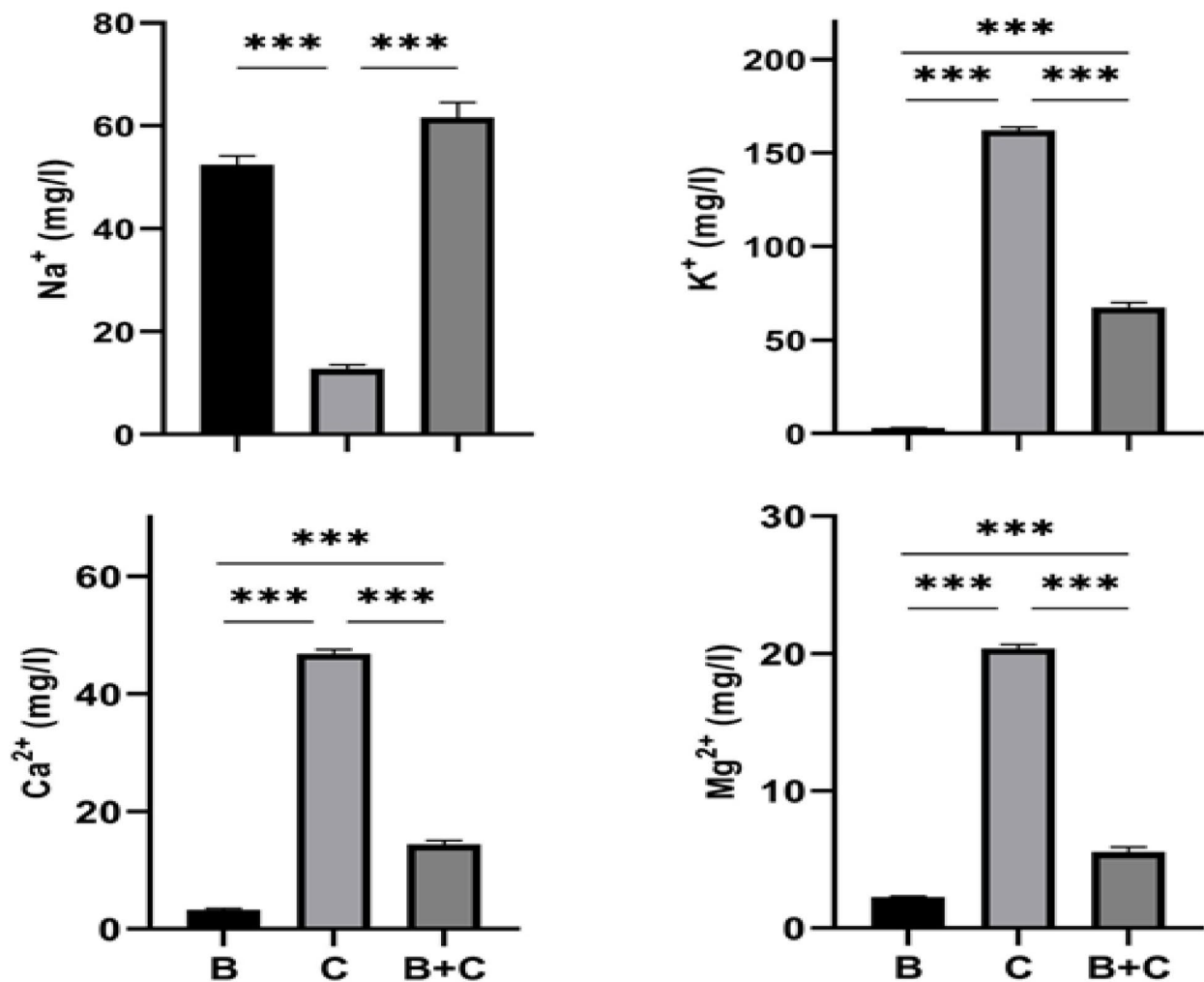
Fig. 3 Cations concentrations at the beginning (0 h); statistic shown: mean $\pm$ SEM ($n = 3$ per treatment); matrix: 0.45 μ m-filtered supernatants; units: mg L⁻¹; treatment means (mg L⁻¹): Na⁺- bentonite [37.88 $\pm$ 6.82], compost [16.45 $\pm$ 5.91], bentonite+compost [40.79 $\pm$ 3.84];

K⁺- bentonite [1.8 $\pm$ 0.26], compost [77.83 $\pm$ 0.26], bentonite+compost [51.84 $\pm$ 6.26]; Ca²⁺ bentonite [5.07 $\pm$ 1.15], compost [45.47 $\pm$ 7.59], bentonite+compost [18.9 $\pm$ 5.64]; Mg²⁺ bentonite [2.98 $\pm$ 0.8], compost [18.76 $\pm$ 2.96], bentonite+compost [8.01 $\pm$ 2.49]

was 12.77 mg L⁻¹ in compost-only, 52.53 mg L⁻¹ in bentonite-only, and 61.64 mg L⁻¹ in bentonite+compost. Both bentonite-containing treatments exceeded compost-only ($p < 0.001$), while the difference between bentonite-only and bentonite+compost remained not significant (n.s.) (Fig. 4).

As shown in Fig. 3, K⁺ concentrations at 0 h were 77.83 mg L⁻¹ in compost-only, 1.80 mg L⁻¹ in bentonite-only, and 51.85 mg L⁻¹ in bentonite+compost. Compost-only exceeded both bentonite-only and bentonite+compost ($p < 0.001$ for both comparisons), and bentonite+compost also exceeded bentonite-only ($p < 0.01$) (Fig. 3). At day 30 (Fig. 4), K⁺ concentrations were 162.11 mg L⁻¹ in compost-only, 2.998 mg L⁻¹ in bentonite-only, and 67.375 mg L⁻¹ in bentonite+compost; all pairwise differences were significant ($p < 0.001$).

Dissolved Ca²⁺ and Mg²⁺ concentrations differed among treatments at both sampling points. At 0 h, compost-only showed the highest concentrations (Ca²⁺ = 45.48 mg L⁻¹; Mg²⁺ = 18.76 mg L⁻¹), significantly exceeding bentonite-only (Ca²⁺ = 5.07 mg L⁻¹, $p = 0.0049$; Mg²⁺ = 2.98 mg L⁻¹, $p = 0.0065$) and bentonite+compost (Ca²⁺ = 18.90 mg L⁻¹, $p = 0.0328$; Mg²⁺ = 8.01 mg L⁻¹, $p = 0.036$). Differences between bentonite-only and bentonite+compost at 0 h were not significant for either Ca²⁺ ($p = 0.2547$) or Mg²⁺ ($p = 0.33$) (Fig. 3). At day 30, compost-only remained highest (Ca²⁺ = 46.80 mg L⁻¹; Mg²⁺ = 20.39 mg L⁻¹) and exceeded both bentonite-only (Ca²⁺ = 3.30 mg L⁻¹, $p = 1.3 \times 10^{-7}$; Mg²⁺ = 2.26 mg L⁻¹, $p = 1.3 \times 10^{-8}$) and bentonite+compost (Ca²⁺ = 14.36 mg L⁻¹, $p = 8.4 \times 10^{-7}$; Mg²⁺ = 5.54 mg L⁻¹, $p =$



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Fig. 4 Cations concentrations at the end (day 30); statistic shown: mean $\pm$ SEM ($n = 3$ per treatment); matrix: 0.45 μ m-filtered supernatants; units: mg L⁻¹; treatment means (mg L⁻¹): Na⁺- bentonite [52.53 $\pm$ 1.64], compost [12.76 $\pm$ 0.82], bentonite+compost [61.64 $\pm$ 2.93]; K⁺-

bentonite [2.99 $\pm$ 0.2], compost [162.11 $\pm$ 1.61], bentonite+compost [5.45 $\pm$ 0.37]; Ca²⁺ bentonite [3.3 $\pm$ 0.25], compost [46.79 $\pm$ 0.8], bentonite+compost [14.36 $\pm$ 0.73]; Mg²⁺ bentonite [2.25 $\pm$ 0.07], compost [20.39 $\pm$ 0.27], bentonite+compost [5.45 $\pm$ 0.37]

4.44 $\times 10^{-8}$); bentonite+compost also exceeded bentonite-only at day 30 for both Ca²⁺ ($p = 0.00046$) and Mg²⁺ ($p = 0.0003$) (Fig. 4). Carbonate-species signal differed among treatments at both sampling points. At 0 h, values were 5.83 mg L⁻¹ in bentonite-only, 6.40 mg L⁻¹ in compost-only, and 12.87 mg L⁻¹ in bentonite+compost; bentonite+compost exceeded bentonite-only ($p < 0.001$) and compost-only ($p < 0.01$), while bentonite-only and compost-only did not differ (n.s.) (Fig. 5). At day 30, values were 7.12 mg L⁻¹ in bentonite-only, 4.93 mg L⁻¹ in compost-only, and 12.83 mg L⁻¹ in bentonite+compost; bentonite+compost exceeded bentonite-only ($p < 0.01$) and compost-only ($p < 0.001$), whereas bentonite-only and compost-only again did not differ significantly (n.s.) (Fig. 6).

bentonite [2.99 $\pm$ 0.2], compost [162.11 $\pm$ 1.61], bentonite+compost [5.45 $\pm$ 0.37]; Ca²⁺ bentonite [3.3 $\pm$ 0.25], compost [46.79 $\pm$ 0.8], bentonite+compost [14.36 $\pm$ 0.73]; Mg²⁺ bentonite [2.25 $\pm$ 0.07], compost [20.39 $\pm$ 0.27], bentonite+compost [5.45 $\pm$ 0.37]

Chloride (Cl⁻) and sulfate (SO₄²⁻) concentrations are shown in Figs. 5 and 6. At 0 h, Cl⁻ concentrations were 358.59 mg L⁻¹ (bentonite-only), 470.29 mg L⁻¹ (compost-only), and 464.41 mg L⁻¹ (bentonite+compost), and no pairwise differences were significant (n.s.) (Fig. 5). By day 30, Cl⁻ concentrations were 241.02 mg L⁻¹ (bentonite-only), 329.20 mg L⁻¹ (compost-only), and 446.78 mg L⁻¹ (bentonite+compost); Cl⁻ was higher in bentonite+compost than bentonite-only ($p < 0.01$) (Fig. 6). For SO₄²⁻, concentrations at 0 h were 20.97 mg L⁻¹ (bentonite-only), 26.65 mg L⁻¹ (compost-only), and 16.82 mg L⁻¹ (bentonite+compost), with no significant pairwise differences (n.s.) (Fig. 5). At day 30, SO₄²⁻ concentrations were 23.71 mg L⁻¹ (bentonite-only), 29.30 mg L⁻¹ (compost-only), and 5.87 mg L⁻¹ (bentonite+compost); compost-only exceeded bentonite-only ($p < 0.01$) and bentonite+compost ($p < 0.001$) (Fig. 6).

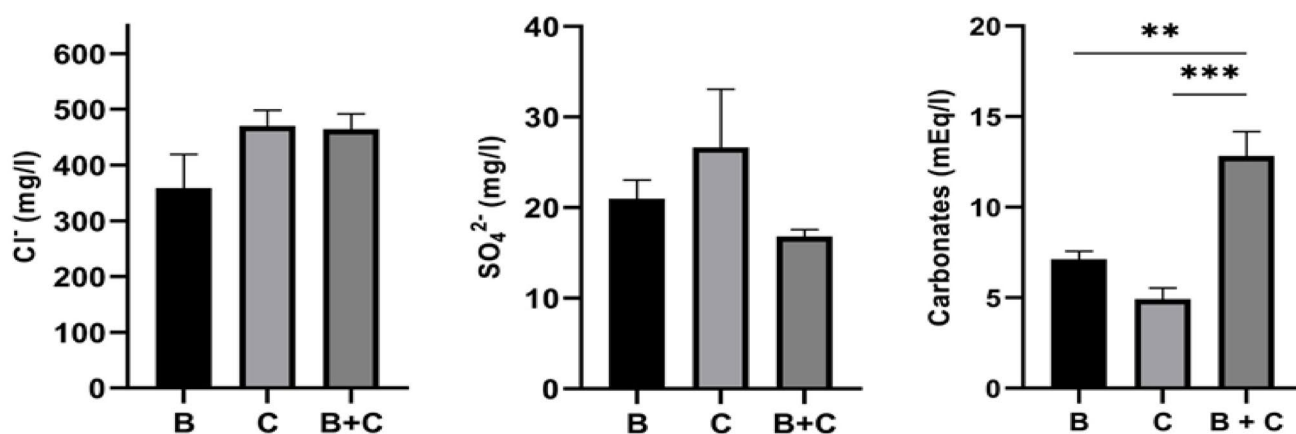


Fig. 5 Anions concentrations at the beginning (0 h); statistic shown: mean $\pm$ SEM ($n = 3$ per treatment); matrix: 0.45 μ m-filtered supernatants; units: mg L^{-1} ; treatment means (mg L^{-1}): Cl^{-} bentonite [358.59 $\pm$ 60.69], compost [470.29 $\pm$ 28.31], bentonite+compost [464.41

$\pm$ 27.94]; SO_4^{2-} bentonite [20.97 $\pm$ 2.07], compost [26.65 $\pm$ 6.41], bentonite+compost [61.82 $\pm$ 0.76]; carbonates- B [5.83 $\pm$ 0.18], C [6.4 $\pm$ 0.64], B+C [12.86 $\pm$ 1.82]

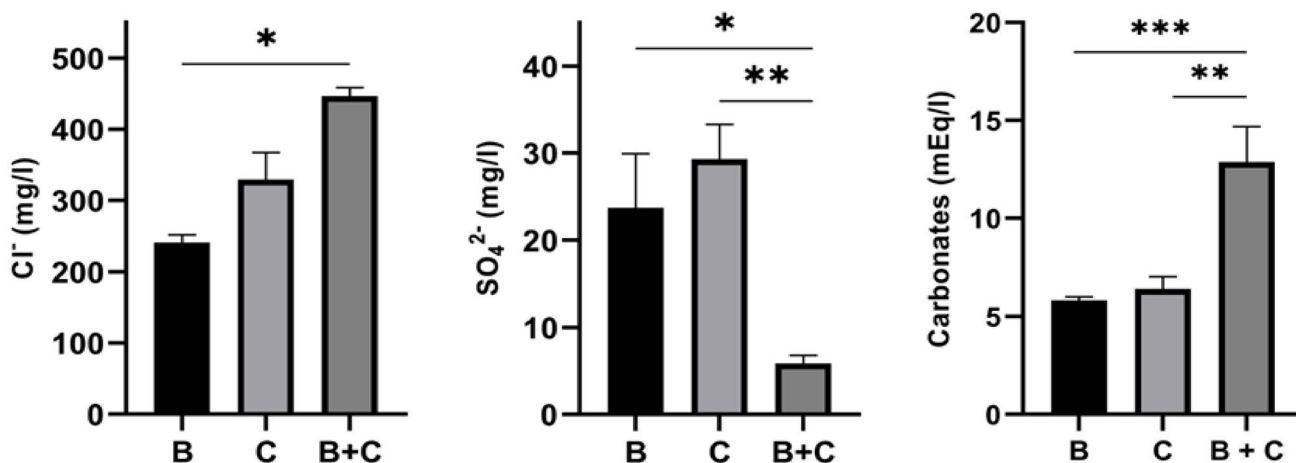


Fig. 6 Anions concentrations at the end (day 30); statistic shown: mean $\pm$ SEM ($n = 3$ per treatment); matrix: 0.45 μ m-filtered supernatants; units: mg L^{-1} ; treatment means (mg L^{-1}): Cl^{-} bentonite [241.02 $\pm$ 10.84], compost [329.20 $\pm$ 38.28], bentonite+compost [446.78

$\pm$ 11.75]; SO_4^{2-} bentonite [23.71 $\pm$ 6.22], compost [29.30 $\pm$ 3.99], bentonite+compost [5.87 $\pm$ 0.91]; carbonates- B [7.11 $\pm$ 0.45], C [4.93 $\pm$ 0.6], B+C [12.83 $\pm$ 1.34]

TN differed among treatments at both sampling points. At 0 h, TN was 2.87 mg L^{-1} in bentonite-only, 13.55 mg L^{-1} in compost-only, and 11.59 mg L^{-1} in bentonite+compost; compost-only and bentonite+compost both exceeded bentonite-only ($p < 0.001$), while compost-only and bentonite+compost did not differ significantly (n.s.) (Fig. 7). At day 30, TN was 1.79 mg L^{-1} in bentonite-only, 28.48 mg L^{-1} in compost-only, and 15.27 mg L^{-1} in bentonite+compost, and all pairwise comparisons were significant ($p < 0.001$) (Fig. 8).

TP differed among treatments at both sampling points. At 0 h, TP was 698 mg L^{-1} in bentonite-only, 7334.33 mg L^{-1} in compost-only, and 1334 mg L^{-1} in bentonite+compost; compost-only exceeded bentonite-only ($p < 0.01$) and bentonite+compost ($p < 0.01$), whereas bentonite-only and bentonite+compost did not differ (n.s.) (Fig. 7). At day 30, TP was 2956.33 mg L^{-1} in bentonite-only, 13,475.5 mg L^{-1} in

compost-only, and 12,966.83 mg L^{-1} in bentonite+compost; compost-only and bentonite+compost both exceeded bentonite-only ($p < 0.001$ for both comparisons), while compost-only and bentonite+compost did not differ significantly (n.s.) (Fig. 8).

TOC differed among treatments at both sampling points. At 0 h, TOC was 3.65 mg L^{-1} in bentonite-only, 67.01 mg L^{-1} in compost-only, and 53.30 mg L^{-1} in bentonite+compost; compost-only exceeded bentonite-only ($p < 0.01$) and bentonite+compost ($p = 0.015$), while bentonite-only and bentonite+compost did not differ (n.s.) (Fig. 7). At day 30, TOC was 6.55 mg L^{-1} in bentonite-only, 174.94 mg L^{-1} in compost-only, and 94.48 mg L^{-1} in bentonite+compost, and all pairwise comparisons were significant ($p < 0.001$) (Fig. 8).

Using the equivalent percentage (mEq%) composition derived from the measured major-ion suite, treatments

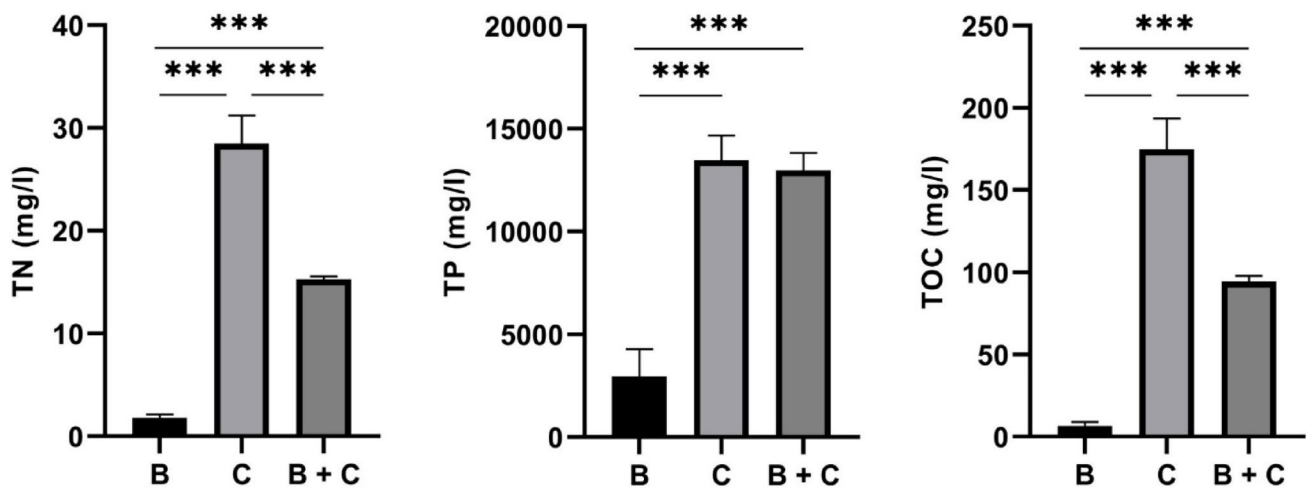


Fig. 7 TN, TP, and TOC concentrations at the beginning (0 h); statistic shown: mean $\pm$ SEM ($n = 3$ per treatment); matrix: $0.45 \mu\text{m}$ -filtered supernatants; units: mg L^{-1} ; treatment means (mg L^{-1}): TN- bentonite [2.86 ± 1], compost [13.54 ± 0.4], bentonite+compost [$11.58 \pm$

0.47]; TP- bentonite [698 ± 163.97], compost [7334.33 ± 1527.25], bentonite+compost [1334 ± 168.53]; TOC- bentonite [3.65 ± 0.9], compost [67.1 ± 10.68], bentonite+compost [53.29 ± 9.95]

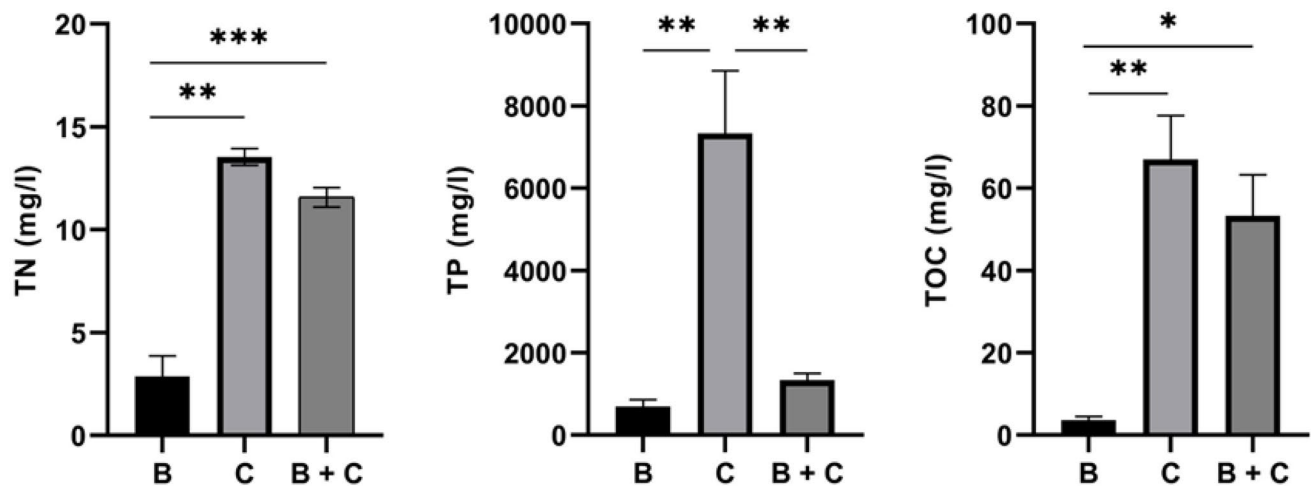


Fig. 8 TN, TP, and TOC concentrations at the end (day 30); statistic shown: mean $\pm$ SEM ($n = 3$ per treatment); matrix: $0.45 \mu\text{m}$ -filtered supernatants; units: mg L^{-1} ; treatment means (mg L^{-1}): TN- bentonite [2.86 ± 1], compost [13.54 ± 0.4], bentonite+compost [11.58 ± 0.47];

TP- bentonite [2956.33 ± 1333.21], compost [13475.5 ± 1200.37], bentonite+compost [12966.83 ± 853.63]; TOC- bentonite [6.54 ± 2.49], compost [174.93 ± 18.74], bentonite+compost [94.47 ± 3.56]

showed clear hydrochemical shifts between the initial (0 h) and final (day 30) sampling points. At 0 h, the bentonite and bentonite+compost treatments were classified as soda-saline (Na-Cl as the dominant pattern, with ($\text{HCO}_3^- + \text{CO}_3^{2-}$) also exceeding 25 e%), whereas the compost-only treatment was classified as saline with a Ca-Cl pattern. By day 30, the bentonite treatment shifted to a $\text{Na-(HCO}_3^- + \text{CO}_3^{2-})$ pattern and was classified as transitional toward soda, while the compost-only treatment remained saline, shifting to a K-Cl pattern. The bentonite+compost treatment also exhibited a $\text{Na-(HCO}_3^- + \text{CO}_3^{2-})$ pattern at day 30 and was likewise classified as transitional toward soda. These equivalents-based classifications are consistent with the observed treatment differences in solute chemistry and support the

interpretation that the combined mineral–organic treatment most strongly promotes movement toward soda-lake-type ionic conditions under the present bottle-incubation setting.

Discussion

In this aqueous bottle system, the pH and EC trajectories reflect the balance between alkalinity-generating mineral–water reactions associated with Na-smectite surfaces and acidifying processes linked to compost decomposition. Smectite-rich clays can buffer pH through edge-site protonation/deprotonation and proton exchange reactions, which together can sustain higher pH and dampen short-term

variability in clay suspensions (Jeon and Nam 2019). In contrast, compost decomposition produces CO_2 and low-molecular-weight organic acids, which increase acidity and can drive pH downward, particularly in treatments where organic matter dominates the reactive pool (Bernal et al. 2009). For carbon stocks, this means that compost addition may both increase dissolved and gaseous carbon release during decomposition and provide organic carbon that can potentially be retained through association with bentonite surfaces. However, this experiment did not directly quantify solid-phase carbon storage or CO_2 flux. The intermediate pH behavior in the combined bentonite+compost treatment is consistent with the co-occurrence of compost-derived acidification and bentonite-associated buffering/alkalinity generation in the same water column. Electrical conductivity (EC) primarily tracks total dissolved ionic strength; thus, higher EC indicates a larger dissolved-ion load arising from the combined contribution of compost-derived solutes (major ions and nutrients released during decomposition) together with bentonite-associated ion partitioning processes (exchange-related redistribution of cations between solid-associated and dissolved pools). Although the observed pH values fall within the alkaline range reported for many soda-lake waters, soda-lake classification is not based on pH alone and must be interpreted together with equivalents-based ion dominance derived from the measured major-ion suite (Boros and Kolpakova 2018). More broadly, these coupled pH–EC responses are consistent with “chemical divide” concepts in alkaline-lake development, where mineralogy, solute acquisition, and hydrologic closure control trajectories toward carbonate-alkaline waters (Tosca and Tutolo 2023).

Treatment differences in evaporation rate likely reflect the contrasting physical behavior of the added solids at the water–solid interface. Bentonite is a swelling clay whose water-association behavior can alter the physical properties of clay–water mixtures, which is consistent with lower evaporation in bentonite-containing bottles (Wang et al. 2022). Compost introduces a coarser particulate matrix and soluble organic compounds that can modify the near-surface properties of the bottle mixture; together, these effects indicate that additive composition can influence evaporation behavior in the experimental system.

The cation patterns show that bentonite-containing treatments were associated with stronger Na enrichment of the aqueous phase than compost-only. In soda-lake biogeochemical evolution, progressive removal of Ca^{2+} and Mg^{2+} through mineral controls and chemical divides favors Na dominance in the residual solution; this framework is widely used to interpret alkaline-lake development in closed basins (Gibbs 1970; Tosca and Tutolo 2023). Consistent with this conceptual model, Na dominance (expressed as equivalents percent) was achieved in the bentonite-only treatment and was

approached in the combined bentonite+compost treatment by the final sampling point, while compost-only did not show Na dominance when evaluated on an equivalent’s basis.

Potassium was strongly associated with compost addition, reflecting the K content of compost and its rapid release to the dissolved phase during incubation. Lower dissolved K in the combined treatment compared with compost-only indicates that adding bentonite altered K partitioning between the filtrate and the bentonite-containing solid fraction. Such treatment-dependent nutrient/ion transformations during co-amendment with bentonite have been reported in composting and amendment studies, consistent with mineral–organic interactions affecting dissolved nutrient pools (Li et al. 2012).

Dissolved Ca^{2+} and Mg^{2+} were highest in compost-only and substantially lower in bentonite-containing treatments, indicating that bentonite presence modified the dissolved divalent-cation fraction in the bottle water. Ca^{2+} and Mg^{2+} contribute disproportionately to charge balance relative to mass; therefore, separating these divalent cations and evaluating their equivalent contributions is essential for robust dominance calculations, as required in soda-lake classifications (Boros and Kolpakova 2018).

Carbonate-system behavior also differed among treatments. The carbonate-species signal was strongest in the combined bentonite+compost treatment, consistent with a larger carbonate-system response under conditions where organic-matter decomposition supplies inorganic carbon to solution and bentonite-containing bottles maintain more alkaline conditions. Interpreted within chemical-divide frameworks, increased carbonate-system dominance relative to sulfate and chloride is a defining feature of alkaline brines and soda-lake waters (Tosca and Tutolo 2023).

Measured chloride and sulfate provide the non-carbonate anion pool against which carbonate-system dominance was evaluated. Chloride typically behaves conservatively during evaporation–concentration and is commonly used as a tracer of evaporative enrichment in closed-basin waters (Gibbs 1970). Sulfate can also behave conservatively over short timescales in oxic waters, but in natural alkaline-lake settings its relative importance may additionally reflect mineral controls (e.g., gypsum/anhydrite) and redox-dependent microbial transformations. Accordingly, including Cl^- and SO_4^{2-} in the equivalents-based anion calculations provides a stricter test of carbonate-system dominance than relying on carbonate measurements alone, aligning the experiment with soda-lake classification practice (Boros and Kolpakova 2018).

Nutrient patterns were dominated by compost addition, consistent with nitrogen and phosphorus inputs from the compost material and release during decomposition. These observations align with the general behavior of organic fertilizers and composts, where mineralization governs the timing and magnitude of dissolved N release (Toner and

Catling 2020; Geisseler et al. 2021). The dissolved organic-carbon signal (TOC) was highest in compost-only and lower in bentonite+compost, indicating that bentonite modified the dissolved/filtered organic-carbon fraction relative to compost alone. Such reductions in dissolved organic matter in the presence of reactive mineral surfaces are consistent with mineral–organic associations that shift organic carbon from the dissolved to solid-associated pool (Kleber et al. 2021). This suggests a potential pathway by which clay–organic matter interactions may contribute to carbon retention; however, direct evaluation of carbon sequestration would require measurements of solid-phase organic carbon, carbonate mineral formation, and CO₂ exchange.

Overall, when evaluated against the operational alkaline soda-lake water criteria (pH threshold and equivalents-based dominance of Na⁺ among cations and carbonate-system species among anions), treatments showed distinct responses. The bentonite-only treatment met the Na-dominance criterion and maintained alkaline pH, while the combined treatment was classified as transitional toward soda based on the equivalents-based major-ion pattern (Na–(HCO₃[−]+CO₃^{2−})) without sustaining alkaline pH continuously over the full incubation; compost-only promoted nutrient and organic-carbon release but did not produce the ionic dominance pattern characteristic of soda-lake waters on an equivalents basis. This separation among treatments is consistent with the broader understanding that soda-lake water chemistry reflects coupled mineral controls, hydrologic closure, and carbon-system evolution rather than pH alone (Tosca and Tutolo 2023).

These bottle experiments are best interpreted as a mechanistic analogue for early-stage brine evolution in closed or weakly flushed settings, where interactions among mineral surfaces, organic inputs, and carbonate-system chemistry can rapidly shift major-ion structure. In applied terms, understanding these short-timescale pathways is relevant for management and rehabilitation of threatened soda-lake wetlands, where hydrologic alteration and drying are increasing pressures in arid and semi-arid regions (Tussupova et al. 2020; Boros and Vörös 2024). Replication was limited, so some differences among treatments may not have been detected.

Conclusions for future biology

This study provides a controlled bottle-scale demonstration that mineral and organic amendments generate distinct solute-chemistry trajectories relevant to the early-stage biogeochemical genesis of alkaline soda-lake water. Compost increased dissolved nutrients and organic carbon in the filtrate, whereas bentonite maintained higher and less variable pH and promoted a Na⁺-dominant cation structure. The combined bentonite+compost treatment showed the

strongest genesis toward soda-lake-type chemistry and was classified as transitional toward soda when evaluated using an equivalents-based framework. These findings support the use of charge-based (meq L^{−1}; e%) classification to interpret early brine evolution and provide mechanistic insight into mineral–organic interactions that are difficult to isolate directly in natural wetlands.

The results are highly relevant not only for understanding the possible pathways involved in the long-term, historical development of pristine soda pans in clay-rich closed-basin landscapes—such as the Pannonian ecoregion—but also for modern, nature-conservation-oriented wetland management. From an applied perspective, the experiment suggests that the interaction between clay-rich substrates and decomposable organic matter, such as plant-derived inputs or livestock droppings, actively can foster sodification processes and helps shift water chemistry toward Na⁺–carbonate-system dominance.

Therefore, the practical conservation significance of this work is substantial. Saline–alkaline soda wetlands are currently highly threatened ecosystems with specialized and endangered biota, and their restoration is often limited by altered hydrology and missing ecological drivers. This study provides crucial baseline experimental evidence to overcome these barriers. Our findings demonstrate that while the restoration of soda pans via river-water flooding is a feasible conservation strategy, it requires careful consideration of both hydrological dynamics and site-management factors to be successful. For wetland rehabilitation, controlled water addition may help restore seasonal inundation and reduce excessive drying, but the biogeochemical outcome will depend strongly on the chemistry of the added water. Compared with the controlled tap-water matrix used here, river water is inherently more variable and may introduce suspended sediments, dissolved organic matter, nutrients, Ca²⁺, Mg²⁺ and bicarbonate, thereby modifying pH, ion dominance, carbonate-system behavior, and the trajectory of brine evolution. Therefore, river-water-based rehabilitation must be evaluated carefully to restore hydrological function while preserving the characteristic Na–carbonate-system chemistry of soda-lake wetlands, which depends on many other environmental factors as well.

Additionally, the observed reduction in dissolved organic carbon in the bentonite+compost treatment suggests that clay–organic matter interactions may support carbon retention by transferring part of the organic carbon from the dissolved phase toward solid-associated pools; however, direct carbon sequestration was not quantified in this experiment. Future research should build on these findings using longer-term mesocosm or field experiments with natural soda-lake sediments and different water sources, including river water and groundwater, to evaluate how mineral–organic

interactions influence brine evolution, carbon retention, habitat quality, and wetland rehabilitation outcomes under more complex environmental conditions.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical approval This is not applicable.

Consent to participate This is not applicable.

Consent for publication This is not applicable.

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